

NOTES & WORKING DRAFTS

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DATE: January 3, 2026 | **PROJECT:** OneDuo™ | **OWNER:** Identity Nails LLC

I. CORE TECHNICAL HYPOTHESIS Existing AI safety measures are "advisory" and can be bypassed via API access. To solve this, governance must be moved to the Persistence Layer.

- **The "Moat" Logic:** If a developer modifies the app to skip a "Submit" button, the database trigger should still catch the transaction and fail it because the human approval signature is missing from the relational record.
- **SQL Constraint as Law:** Use a Boolean gate where state transition evaluates to TRUE only upon verification of an immutable approval record.

II. WORKING DRAFT: INTENT RECONSTRUCTION SPECS We need a passive way to prove the AI acted on human intent without constant recording.

- **Sampling Rate:** Utilizing a predetermined sampling interval sufficient to capture meaningful delta in human creative emphasis.
- **Matching Logic:** Utilize perceptual similarity metrics. A confidence threshold ensures the AI's "reasoning" maps to the frame the human was focused on.
- **The "Reasoning Ledger":** Must be Append-Only. No one can change what the AI "thought" once the human has provided a disposition.

III. WORKING DRAFT: FORENSIC PURGE PROTOCOL Privacy is a product feature. We must prove we deleted the source content while keeping the artifact valid.

- **Step 1 (Hash):** Canonicalize the data and generate a cryptographic fingerprint.
- **Step 2 (Bind):** Store that hash in the audit log and link it to the artifact.

- **Step 3 (Purge):** Trigger the physical deletion event.

IV. DESIGN-AROUND DEFENSES (FOR COUNSEL)

- **Ghost-Path Protection:** During reduction-to-practice testing, we identified that triggers must fire on all state-bearing operations (INSERT, UPDATE, JSONB metadata) and cover secondary paths like materialized views, server-side functions, and replicated tables.
- **The "Scarlet Letter" (Claim 5):** Injected violation watermarks provide a "Dye Pack" forensic trail if a bypass is attempted.